

Physics-Informed Machine Learning (PIML) for Solving Partial Differential Equations

Presenter:

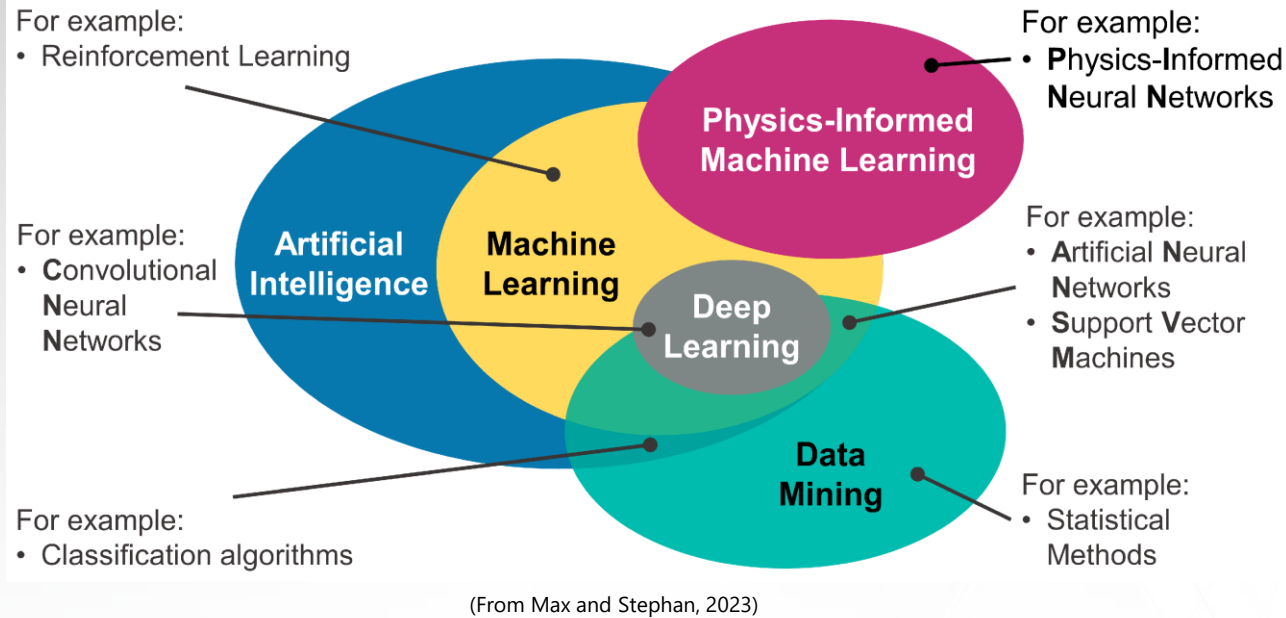
- Lal Mamud
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- **Mentors:** Maruti K. Mudunuru, Satish Karra

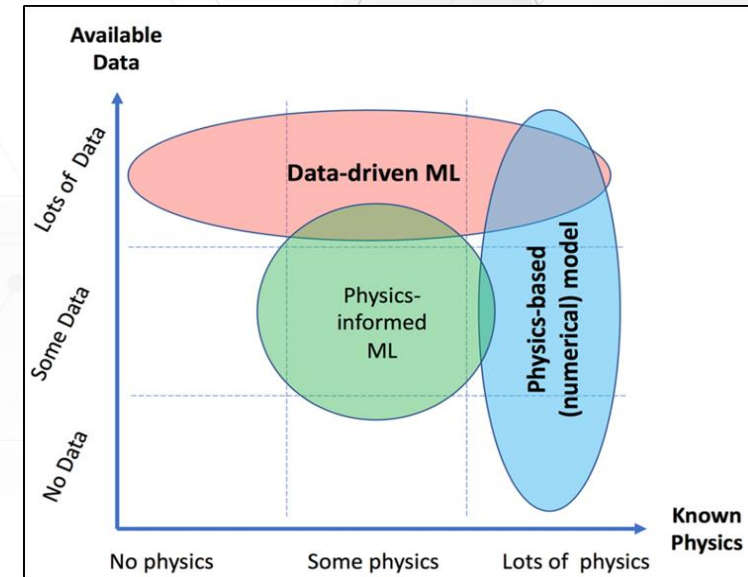
Outline of this Presentation

- What is Physics-Informed Machine Learning (PIML)?
- How does PIML solve partial differential equation (PDE) problems?
- Why PIML:
 - Advantages
 - Limitations
- Examples from the literature.
- Overview of the PDE problems:
 - Demonstrations
 - Exercise
- How the workshop will be conducted.
- What are the expected outcomes of this workshop?

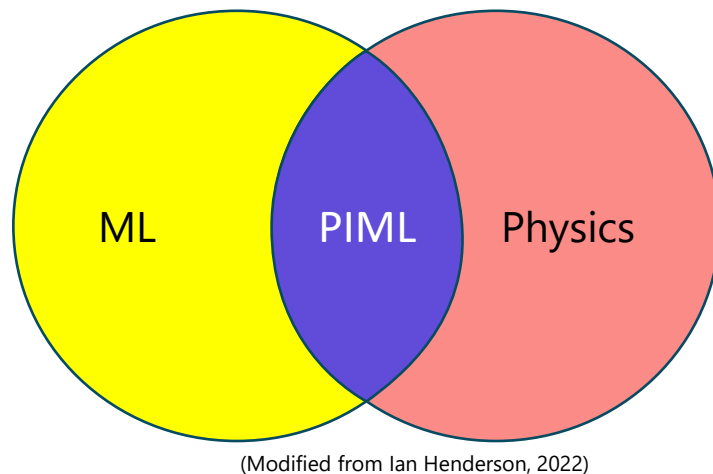
Physics-Informed Machine Learning (PIML)



- **PIML** is a neural network that is trained to respect given laws of physics and/or data.



Data-driven vs. physics-based problems (from Tartakovsky et al. 2020).

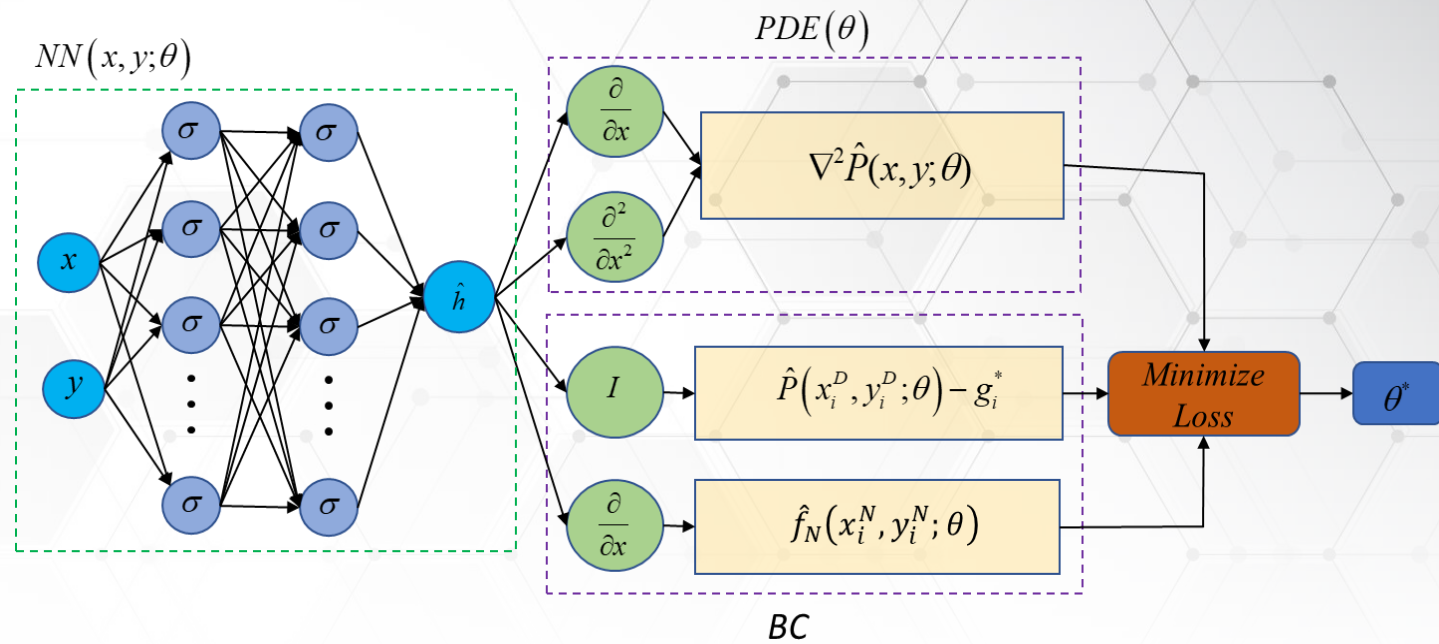


- **Key Features:**

- PINN solves ODEs and PDEs with BCs and ICs.
- PINN incorporates physics equations through its loss function.
- Leverages modern optimization techniques and auto-differentiation.
- It can include data for better simulation and parameter estimation.
- Handles problems in diverse domains like fluid flow, heat transfer, and wave propagation.

Steps to solve PDE problems with PIML

- Domain, BCs, and ICs:** define domain and boundary conditions and generate collocation Points.
- PDE Residual:** Discretize PDEs using auto-differentiation.
- Neural Network:** Create a function to initialize neural network parameters (weights and biases).
- Loss Function:** Define a composite loss function combining residuals from PDEs and boundary and/or initial conditions.
- Hyperparameters and Training Loop:** Define hyperparameters, create parameters update function, and design a training loop.
- Prediction:** Predict the variable of interest with the trained PIML model.



Data-driven PIML architecture for solving the 2D Steady State Problem

$$Loss(\theta) = \underbrace{\frac{1}{N_c} \sum_{i=1}^{N_c} \left[\hat{f}(x_i^c, y_i^c; \theta) \right]^2}_{PDE} + \underbrace{\frac{1}{N_N} \sum_{i=1}^{N_N} \left[\hat{f}_N(x_i^N, y_i^N; \theta) \right]^2}_{Neuman\ BC} + \underbrace{\frac{1}{N_D} \sum_{i=1}^{N_D} \left[\hat{P}(x_i^D, y_i^D; \theta) - g_i^* \right]^2}_{Dirichlet\ BC}$$

$$Loss = PDE_{loss} + BC_{loss} + Data_{loss} + Others_{loss}$$

$$\text{Just NN: } Loss = Data_{loss}$$

Why PIML? Advantages and Limitations

PIML is a transformative approach to solve PDEs by combining physics knowledge and machine learning power.

▪ **Advantages:**

- Mesh-free approach to solving PDEs.
- Flexibility in handling irregular geometries and multi-dimensional problems.
- Once a PIML tuned and trained, it can predict variable of interest very fast.
- Unified framework for solving forward and inverse problems without traditional discretization.
- Suitable for solving nonlinear, convection-dominated, and shock problems.

▪ **Limitations:**

- Computationally expensive training process.
- Sensitivity to hyperparameters such as learning rate, epoch, number of layers and collocation points

While PIML offers unprecedented flexibility and capabilities, understanding its limitations is crucial for effective application.

Examples-1 from Literature: Solving nonlinear PDEs

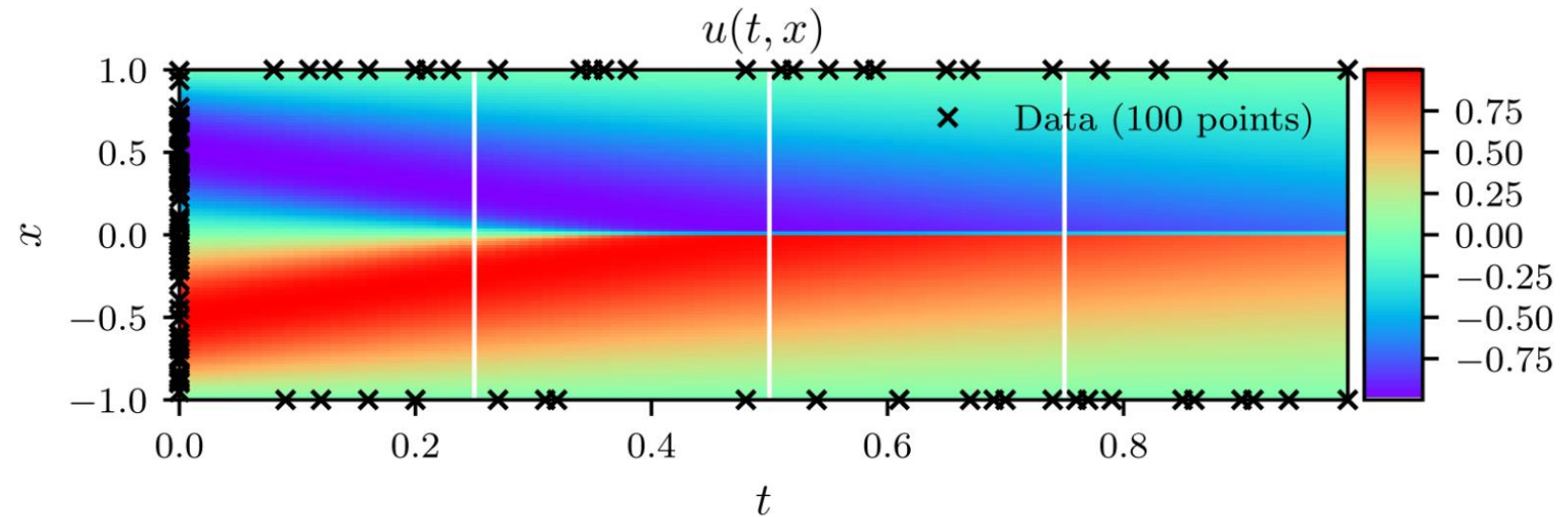
Viscous Burgers' Equation:

- Fundamental example of a parabolic partial differential equation with both nonlinear (convective) and linear (diffusive) terms.
- Can be used in Fluid Dynamics, Shock Wave Formation, Traffic Flow, and Turbulence Modeling

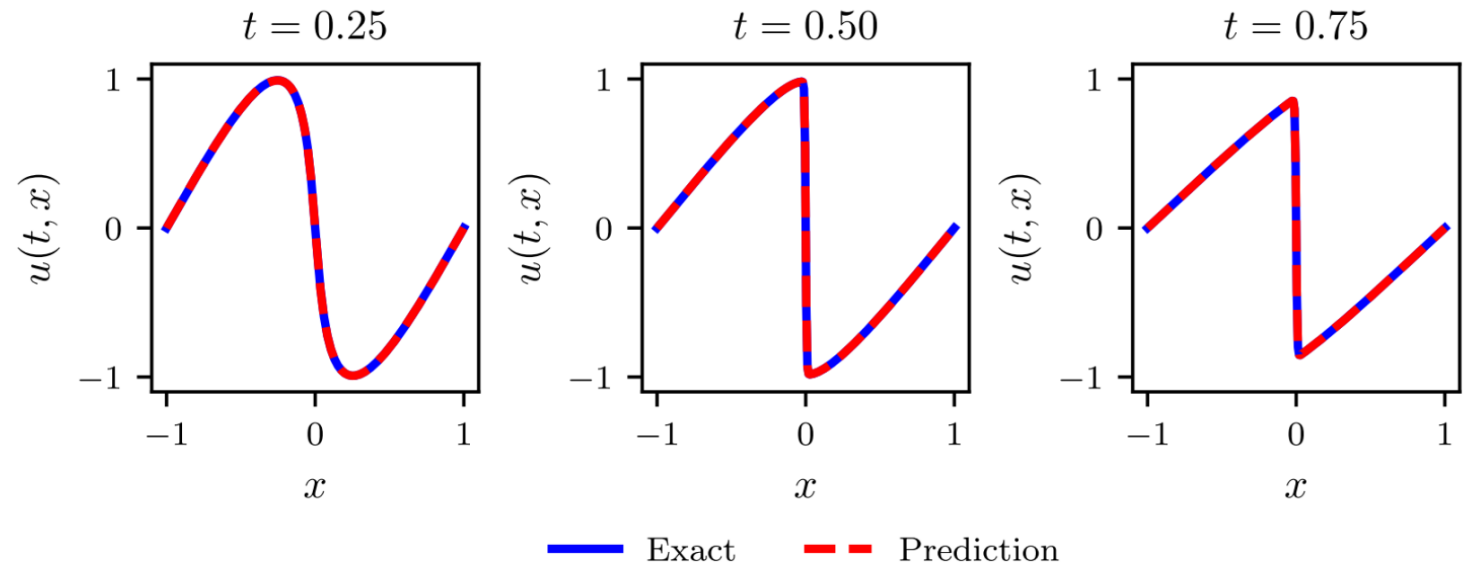
$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} - \frac{0.01}{\pi} \frac{\partial^2 u}{\partial x^2} = 0,$$

$$u(0, x) = -\sin(\pi x),$$

$$u(t, -1) = u(t, 1) = 0.$$



PIML predicted solution, $u(t, x)$ (from Raissi et al., 2017)



Comparison of the predicted and exact solutions corresponding to the three temporal snapshots (from Raissi et al., 2017)

Examples-2 from Literature: Solving nonlinear PDEs

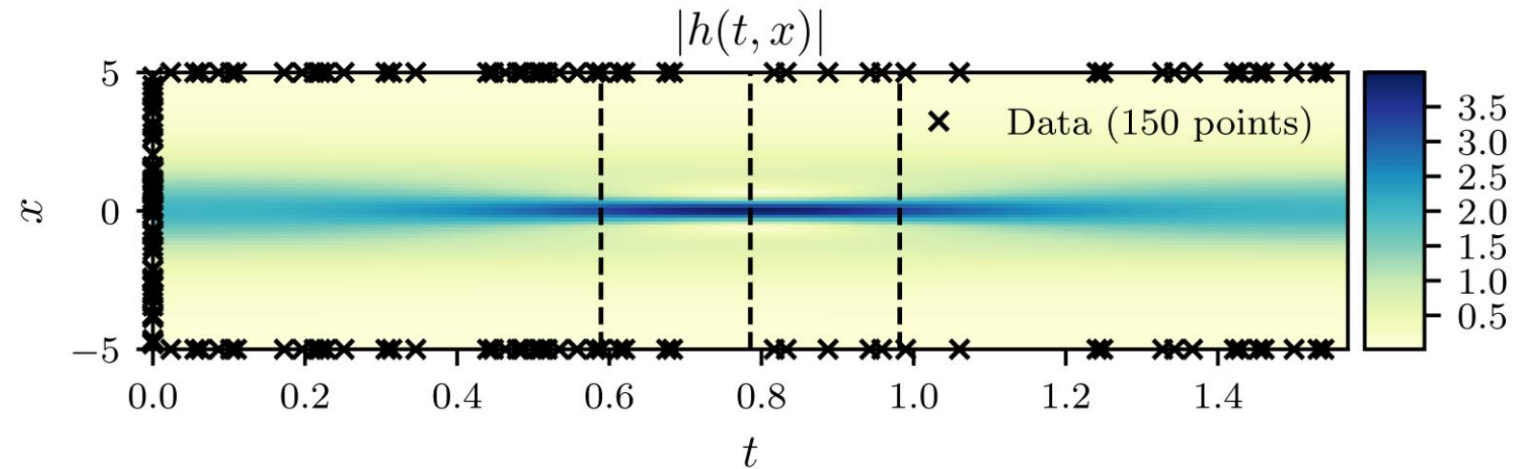
Nonlinear Schrödinger Equation:

- Can be used in Fluid Dynamics, Plasma Physics, Quantum Mechanics and Optics.

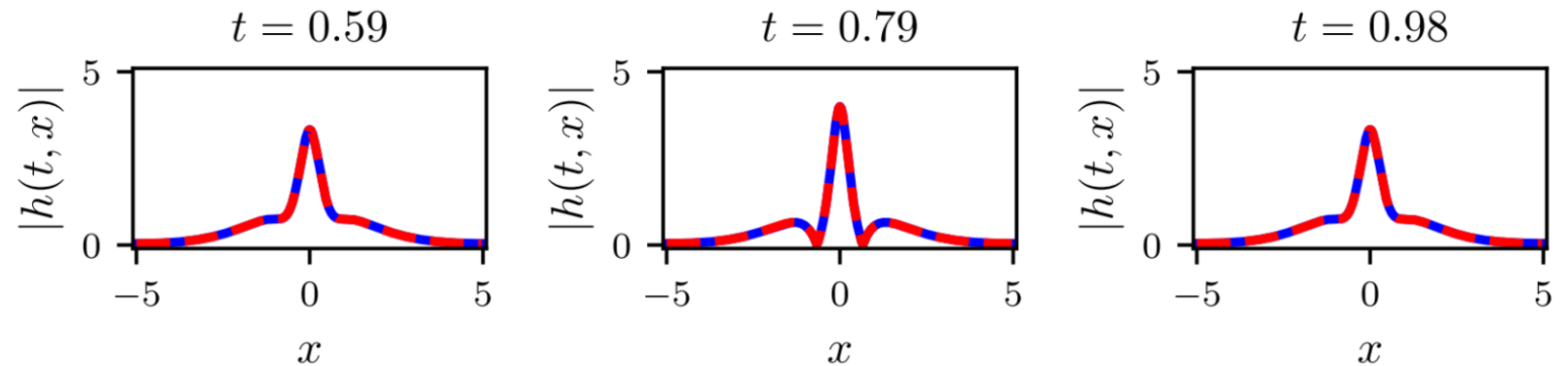
$$i \frac{\partial h}{\partial t} + 0.5 \frac{\partial^2 h}{\partial x^2} + |h|^2 h = 0,$$

$$h(0, x) = 2 \operatorname{sech}(x),$$

$$h(t, -5) = h(t, 5), \quad h_x(t, -5) = h_x(t, 5),$$



PIML predicted solution, $h(t, x)$ (from Raissi et al., 2017)



— Exact - - - Prediction

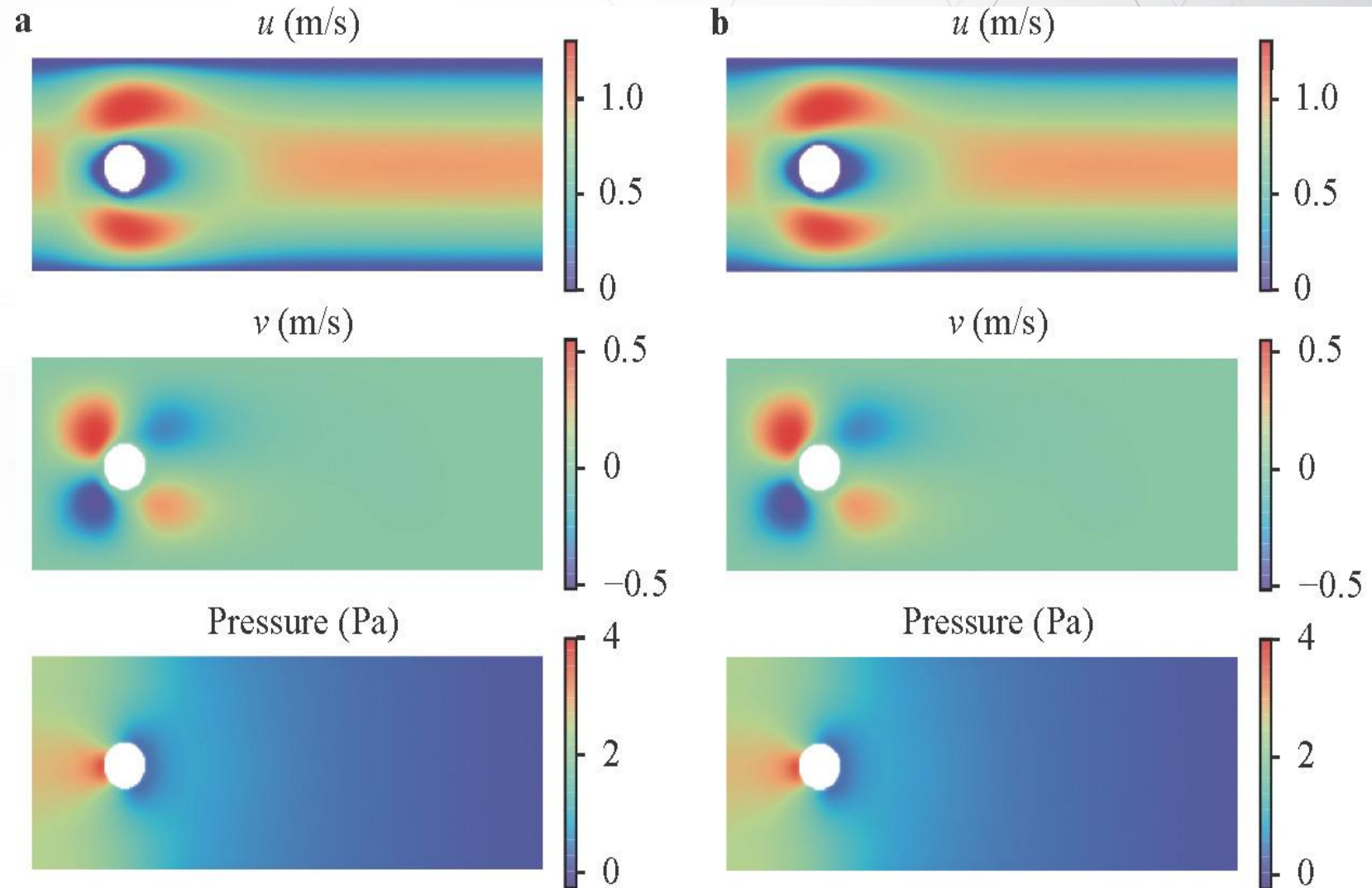
Comparison of the predicted and exact solutions corresponding to the three temporal snapshots (from Raissi et al., 2017)

Examples-3 from Literature: Use of PIML for complex fluid dynamics problems

Navier–Stokes equations:

- Steady-state flow velocity and pressure fields simulation with PIML
- Can be used in aerospace engineering, biomedical engineering, weather and climate Science.

$$(\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{1}{\rho} \nabla p + \frac{\mu}{\rho} \nabla^2 \mathbf{v} + \mathbf{g},$$
$$\nabla \cdot \mathbf{v} = 0,$$



Velocity and pressure fields of the steady flow passing a circular cylinder. a) Reference solution from ANSYS Fluent and b) PIML solution (From Rau et al., 2020).

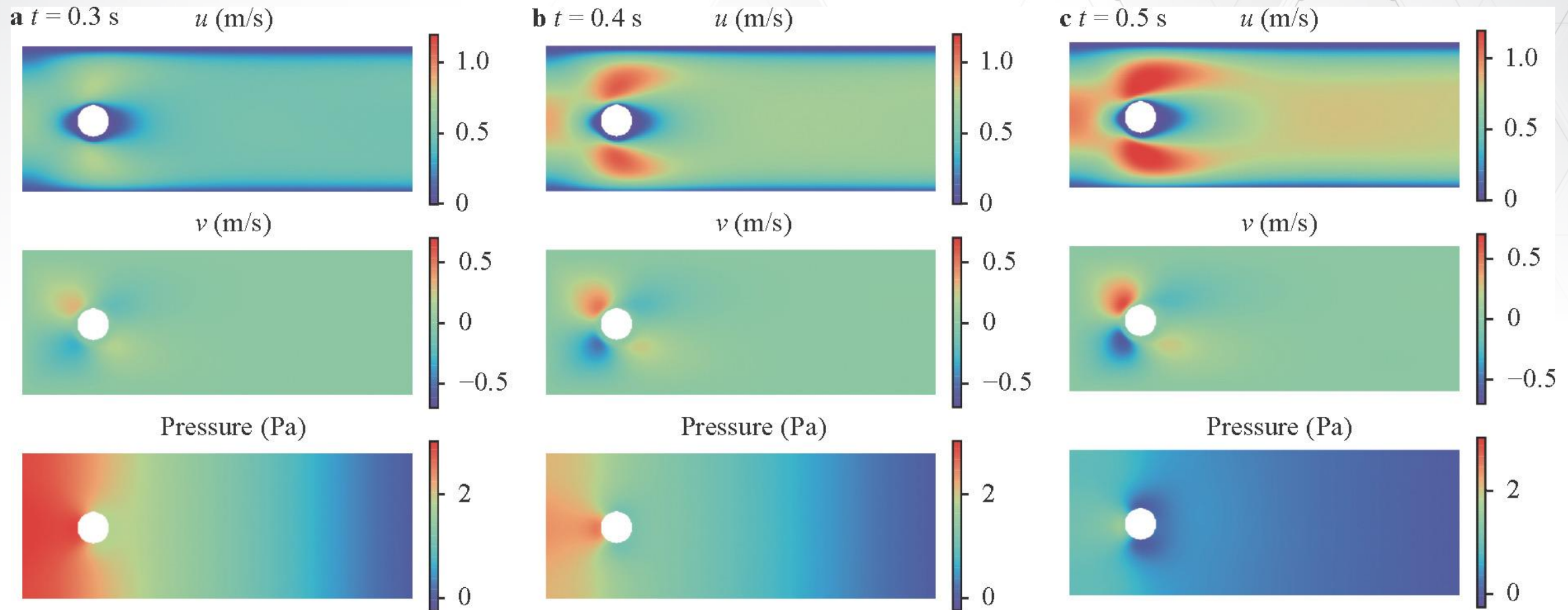
Examples-4 from Literature: Use of PIML for complex fluid dynamics problems

▪ Navier–Stokes equations:

- Transient-state flow velocity and pressure fields simulation with PIML

$$\nabla \cdot \mathbf{v} = 0,$$

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{1}{\rho} \nabla p + \frac{\mu}{\rho} \nabla^2 \mathbf{v} + \mathbf{g},$$



Three snapshots of the predicted flow fields (From Rau et al., 2020)

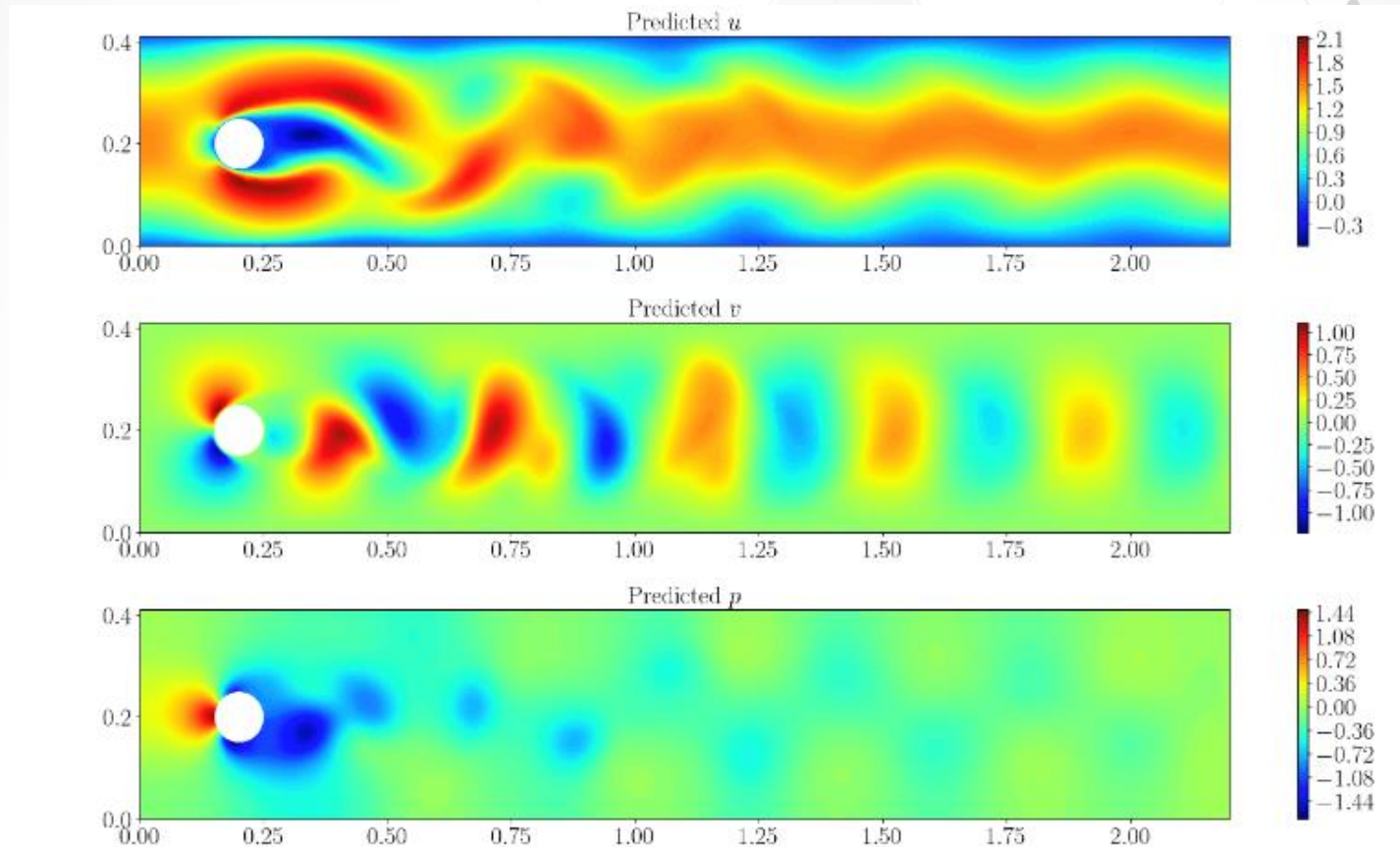
Examples-5 from Literature: Use of PIML for complex fluid dynamics problems

▪ Navier–Stokes equations:

- Transient-state flow velocity and pressure fields simulation with PIML

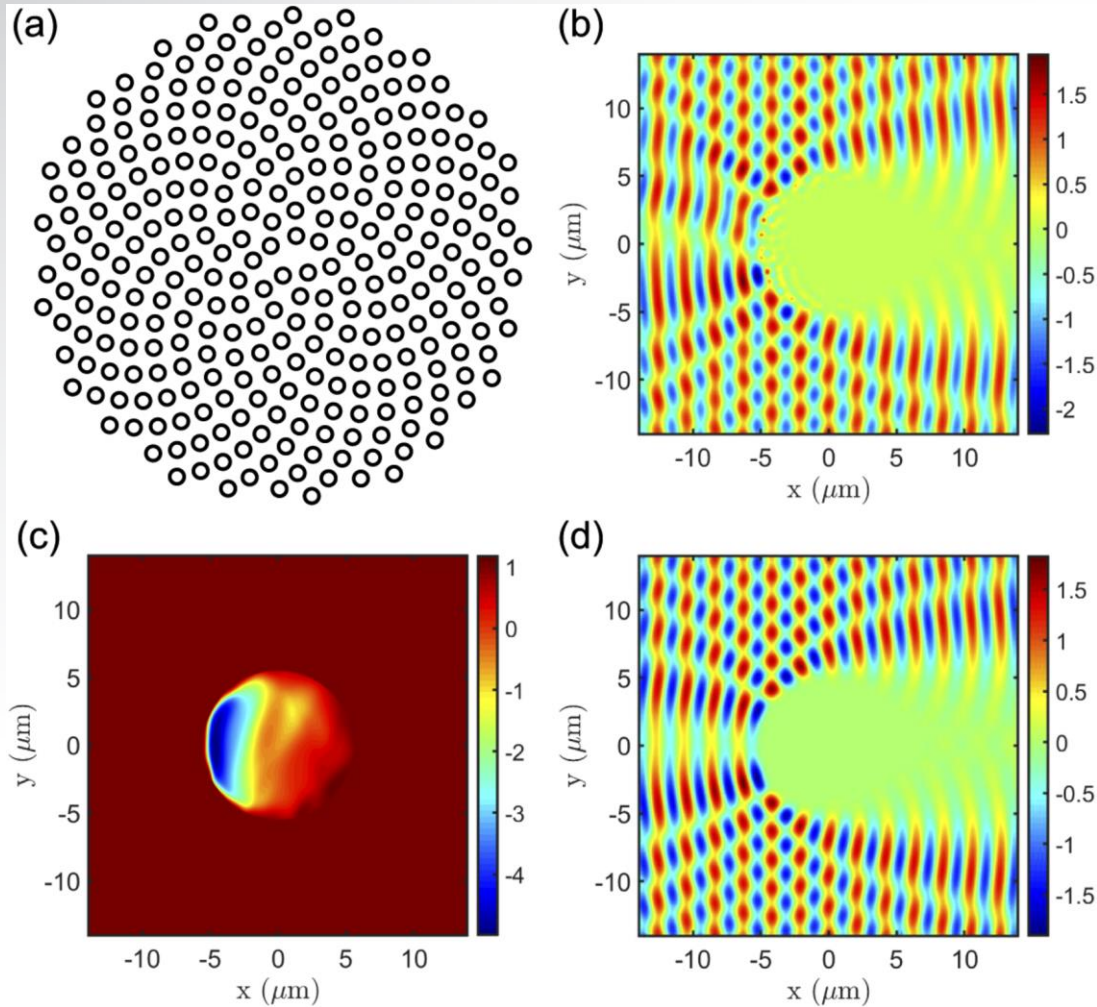
$$\nabla \cdot \mathbf{v} = 0,$$

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{1}{\rho} \nabla p + \frac{\mu}{\rho} \nabla^2 \mathbf{v} + \mathbf{g},$$

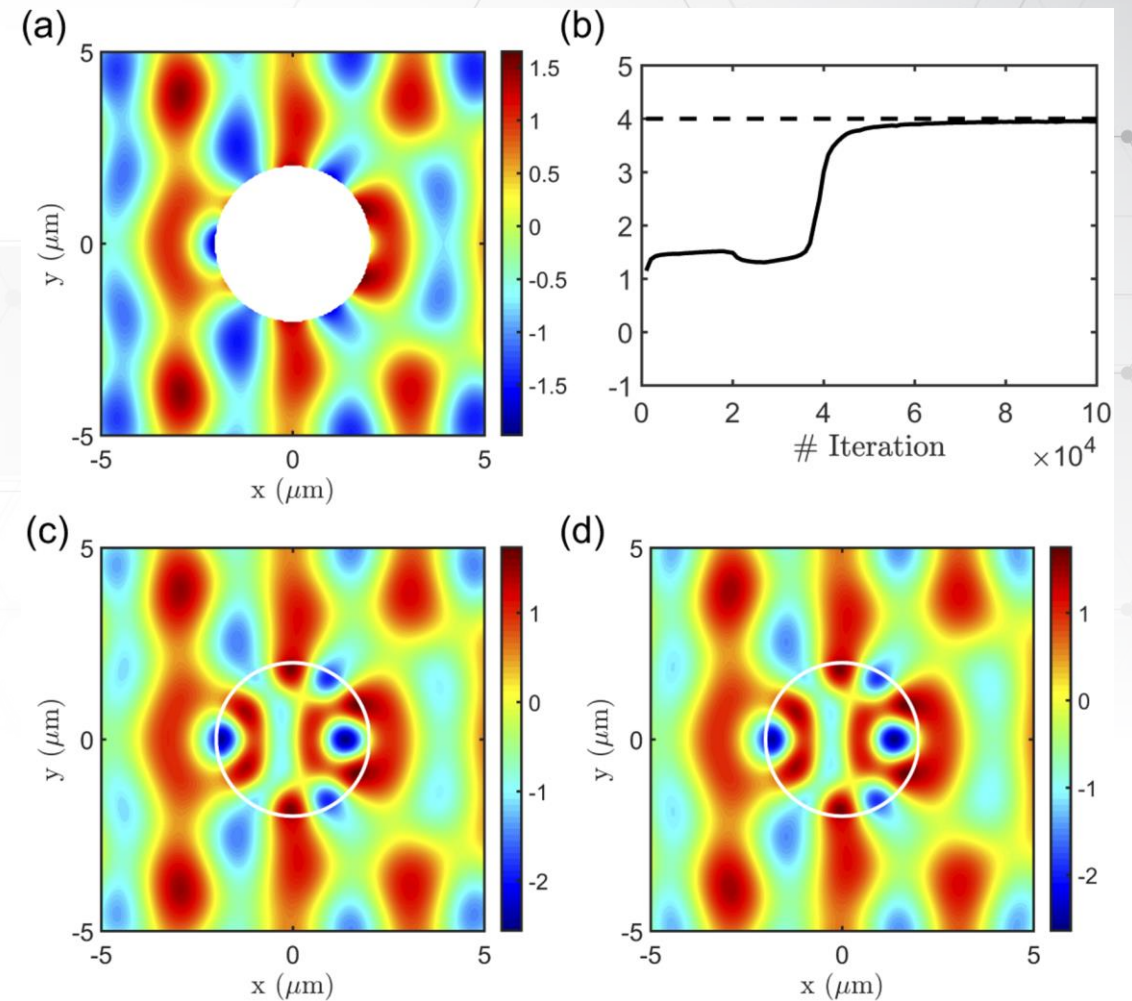


Predicted velocity field and pressure field for $Re = 100$ (Wang et al. 2023)

Examples-6 from Literature: Inverse problems



(a) FEM simulation of the real part of the electric field distribution for a Vogel spiral with $\lambda=2.1 \mu m$, (b) Inverse dataset for PINNs training, derived from panel (a), (c) PINNs-predicted inverse permittivity profile trained using panel (b), (d) FEM validation of the electric field using the PINNs-predicted permittivity. The L_2 -error between panels (b) and (d) is 3.8% (Chen et al., 2020).



(a) Training dataset for PINNs inverse Mie scattering with $r=2 \mu m$, $\epsilon_r=4$, and $\lambda=3 \mu m$, (b) PINNs-predicted nanocylinder permittivity over iterations, (c) Reconstructed electric field distribution from PIML, and (d) Real electric field distribution using Mie theory. The L_2L_2 -error between panels (c) and (d) is 0.51% (Chen et al., 2020).

Literature showcases the versatility of PIML in tackling diverse problems effectively.

Overview of PDE Problems demonstration in this Workshop

1. Darcy's 1D homogeneous flow

$$\frac{\partial^2 h}{\partial x^2} = 0, \quad \text{in } \Omega = \{x \mid 0 \leq x \leq L\}$$
$$h(0) = h_1, \quad h(L) = h_2.$$

2. 2D Darcy's flow homogeneous flow

$$\frac{\partial^2 P}{\partial x^2} + \frac{\partial^2 P}{\partial y^2} = 0, \quad \text{in } \Omega = \{(x, y) \mid 0 \leq x \leq L_1, 0 \leq y \leq L_2\}$$

$$P(0, y) = P_1, \quad \forall y \in [0, L_2].$$

$$P(L_1, y) = P_2, \quad \forall y \in [0, L_2].$$

$$\frac{\partial P}{\partial y}(x, 0) = 0, \quad \forall x \in [0, L_1].$$

$$\frac{\partial P}{\partial y}(x, L_2) = 0, \quad \forall x \in [0, L_1].$$

3. Exercise Problem,

Darcy's 1D heterogeneous flow

$$\frac{\partial}{\partial x} \left(K(x) \frac{\partial h}{\partial x} \right) = 0, \quad \text{in } \Omega = \{x \mid 0 \leq x \leq L\}$$

$$h(0) = h_1, \quad h(L) = h_2.$$

4. Wave Equation

$$u_{tt} = c^2 u_{xx}, \quad x \in [0, \pi], \quad t \in [0, 2\pi]$$

$$u(0, t) = 0, \quad u(\pi, t) = 0,$$

$$u(x, 0) = \sin(x), \quad u_t(x, 0) = c \sin(x),$$

5. Advection Equation

$$u_t + cu_x = 0, \quad x \in [0, 2\pi], \quad t \in [0, 1]$$

$$u(0, x) = \sin(x)$$

$$u(t, 0) = u(t, 2\pi)$$

6. Allen-Cahn Equation

$$u_t - \lambda_1 u_{xx} + \lambda_2 u^3 - \lambda_2 u = 0, \quad x \in [-1, 1], \quad t \in [0, 1]$$

$$u(0, x) = x^2 \cos(\pi x)$$

$$u(t, -1) = u(t, 1), \quad u_x(t, -1) = u_x(t, 1)$$

$$\text{where } \lambda_1 = 0.0001 \text{ and } \lambda_2 = 5.$$

By practicing real PDE problems, participants will develop hands-on expertise in implementing PIML techniques.

How the Workshop Will Be Conducted

■ Pre-Workshop Preparation:

- Instructions for setting up Python environment with required packages.
- Course materials.

■ Workshop Agenda: hands-on session with interactive coding and demonstrations.

- ✓ **8:30 – 9:00:** 30-minute presentation session, question and answer.
- **9:30 - 9:30:** 30-minute installation troubleshooting.
- **9:30 – 10:15:** Demonstration.
- **10:15 – 10:30:** *Coffee Break.*
- **10:30 – 12:00:** Demonstration.
- **12:00 – 1:00:** *Launch*
- **1:00 – 2:45:** 90-minute Hands-on exercise.
- **2:45: – 3:00:** *Coffee Break.*
- **3:00 – 5:00:** Demonstration.

Expected Outcomes of the Workshop

▪ **By the End of the Workshop, Participants Will:**

- Understand the fundamentals of PIML and its applications.
- Formulate PDE problems as machine learning tasks.
- Gain practical skills in using Python for solving PDEs.
- Be equipped to apply PIML techniques to their research problems.

▪ **Networking Opportunity:**

- Collaborate with peers and
- National Laboratory researchers.

Participants will leave with the knowledge, skills, and connections to advance their research using PIML.

Questions?

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